

Quiz (SP 2026)

Class: BSE-4A

Subject: Artificial Intelligence

Time: 10 Minutes

Total Marks: 16

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Registration # FA24-BLE



	Answers Part-I					Answers Part-II			
Q01	☒	☐ B	☐ C	☐ D	Q01	☒	☐ B	☐ C	☐ D
Q02	☐ A	☒	☐ C	☐ D	Q02	☒	☐ B	☐ C	☐ D
Q03	☐ A	☐ B	☒	☐ D	Q03	☒	☐ B	☐ C	☐ D
Q04	☐ A	☐ B	☐ C	☒	Q04	☒	☐ B	☐ C	☐ D
Q05	☐ A	☒	☐ C	☐ D	Q05	☒	☐ B	☐ C	☐ D
Q06	☐ A	☐ B	☒	☐ D	Q06	☒	☐ B	☐ C	☐ D
Q07	☐ A	☐ B	☒	☐ D	Q07	☒	☐ B	☐ C	☐ D
Q08	☐ A	☐ B	☒	☐ D	Q08	☒	☐ B	☐ C	☐ D

Part-I

1. Comparing BFS, UCS, Greedy, and A* by the criterion used for node selection, which ordering is correct?

A. BFS: depth; UCS: $g(n)$; Greedy: $h(n)$; A*: $g(n)+h(n)$	B. BFS: $h(n)$; UCS: $g(n)+h(n)$; Greedy: $g(n)$; A*: depth
C. BFS: $g(n)$; UCS: depth; Greedy: $g(n)+h(n)$; A*: $h(n)$	D. BFS: $g(n)+h(n)$; UCS: $h(n)$; Greedy: depth; A*: $g(n)$

2. In the optimality proof of A*, the inequality $f(n) \leq f(G_0)$ is derived from $h(n) \leq h^*(n)$. This relies on which property?

A. Consistency (monotonicity) of h	B. Monotonicity: f values never decrease along any path
C. $h(\text{goal}) = 0$ for every goal node	D. Admissibility: h never overestimates the true remaining cost h^*

3. A consistent (monotone) heuristic satisfies $h(n) \leq c(n, a, n') + h(n')$. This condition is also known as the:

A. Admissibility condition	B. Manhattan constraint
C. Optimistic bound condition	D. Triangle inequality for heuristics

4. A DFS on a tree with branching factor $b=4$ and max depth $m=10$ pursues a path to maximum depth. Its peak space requirement is:

A. $O(bm)$: stores only the current path plus unexpanded siblings — about $b \times m$ nodes	B. $O(b^m) = 4^{10} \approx 1$ million nodes
C. $O(b^d)$ where d is the solution depth	D. $O(1)$ since DFS discards explored subtrees immediately

5. Why is the straight-line distance (SLD) an admissible heuristic for road navigation problems?

A. The actual road distance always equals the straight-line distance	B. It overestimates, making A* more conservative
C. It is consistent by the triangle inequality of roads	D. The straight-line distance is a lower bound — actual travel is always at least as long

6. Which uninformed search is both complete and optimal on graphs with non-negative, non-zero step costs, and why?

A. BFS — because it finds the shallowest goal	B. DFS — because it explores deepest nodes first
C. UCS — because it expands in order of increasing path cost	D. IDS — because it combines BFS completeness and DFS space efficiency

7. Heuristic h_2 dominates h_1 ($h_2 \geq h_1$ everywhere) and both are admissible. Using h_2 in A* instead of h_1 :

A. Always expands strictly fewer nodes — never the same count	B. May expand more nodes because h_2 values are larger
C. Guarantees a strictly shorter solution path	D. A^* with h_2 expands a subset of nodes that A^* with h_1 would expand (h_2 prunes the search space more aggressively)

8. The Expectiminimax algorithm generalizes minimax to stochastic games. A chance node's value is computed as:

A. max over all possible outcomes	B. min over all possible outcomes
C. $\sum P(\text{outcome}) \times \text{Expectiminimax}(\text{outcome})$ — a probability-weighted average	D. The Expectiminimax value of the most likely outcome

Part-II

1. Depth-limited search (DLS) is preferred over IDS when:

A. The solution depth d is unknown	B. The state space is infinite
C. Memory is the primary constraint	D. An upper bound L on the solution depth is known from domain knowledge

2. IDS and BFS share the same asymptotic time complexity $O(b^d)$. Yet IDS regenerates upper nodes multiple times. Why is IDS still preferred over BFS in large search spaces?

A. IDS always finds the solution faster	B. IDS is optimal while BFS is not
C. IDS has linear $O(bd)$ space whereas BFS requires $O(b^d)$ space	D. IDS uses a heuristic to guide the search

3. Minimax time complexity is $O(b^m)$. For Chess ($b \approx 35$, $d \approx 100$), this is approximately 10^{154} . This makes exhaustive minimax:

A. Feasible with modern parallel hardware	B. Completely infeasible; heuristic evaluation with depth cutoffs is required
C. Feasible only for endgame positions with ≤ 8 pieces	D. Feasible using alpha-beta pruning alone

4. UCS expands nodes in order of increasing path cost $g(n)$. Which fringe data structure enables this efficiently?

A. FIFO queue (BFS order)	B. LIFO stack (DFS order)
C. Min-heap (priority queue ordered by $g(n)$)	D. Hash table ordered by node depth

5. Move ordering in alpha-beta critically affects performance. If the worst moves are always explored first (worst-case ordering), the resulting complexity is:

A. $O(b^{(m/2)})$ — same improvement as ideal ordering	B. $O(b^m)$ — identical to plain minimax, no pruning benefit at all
C. $O(b^{(m/3)})$ — moderate improvement	D. $O(2^m)$ — exponentially worse than minimax

6. In A^* search, nodes are expanded in non-decreasing order of $f(n)$. This means A^* is effectively searching layers of increasing:

A. Node depth	B. Heuristic value $h(n)$ alone
C. Combined cost $f(n) = g(n) + h(n)$	D. Path cost $g(n)$ alone

7. DFS and IDS are both non-optimal in general. Under which specific condition does IDS become optimal?

A. When all step costs are equal (unit cost)	B. When the branching factor is finite
C. When a consistent heuristic is applied	D. When the depth limit equals the maximum tree depth

8. Minimax performs a depth-first search of the game tree. Why is DFS chosen over BFS for minimax?

A. DFS finds the optimal move faster than BFS	B. DFS requires $O(bm)$ space vs. BFS $O(b^m)$, making it feasible
C. BFS cannot propagate minimax values back to the root	D. DFS is complete while BFS is not for game trees